

Anatomy of the foot

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2025



Objectives

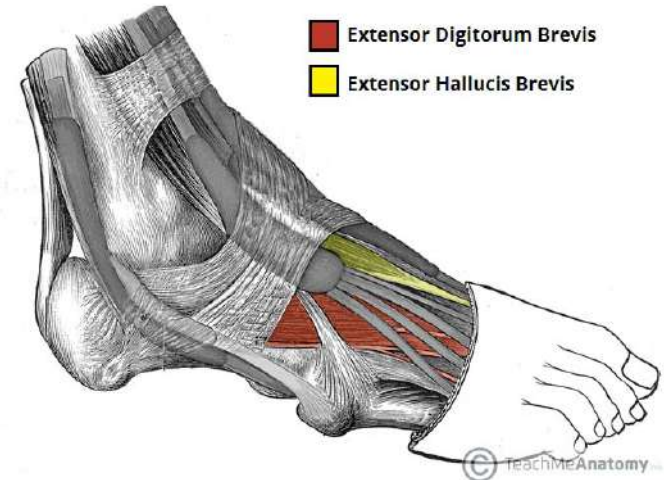
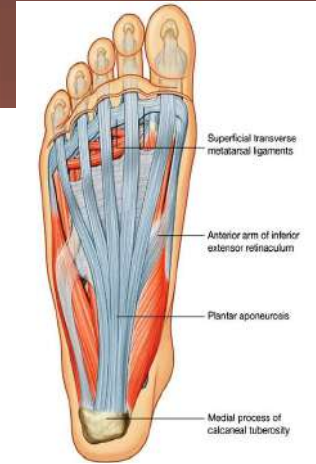
❖ Skeleton of the foot

❖ Sole of the foot

- Skin
- Deep fascia
- Muscles
- Arteries
- Veins
- Nerves

❖ Dorsum of the foot

- Skin
- Dorsal venous arch
- Muscles
- Artery
- Nerves



Foot skeleton

1) Tarsal bones

Are formed of 7 bones: (2)proximal, (1)intermediate
(4)distal.

Proximal

- **Calcaneum**: is the largest bone, forms the prominence of heel, articulates above with the talus, in front with cuboid & has 6 surfaces.
- **Talus**: articulates above at the ankle joint with the tibia & fibula and below with the calcaneum.

Intermediate

- **Navicular**: has tuberosity

Distal

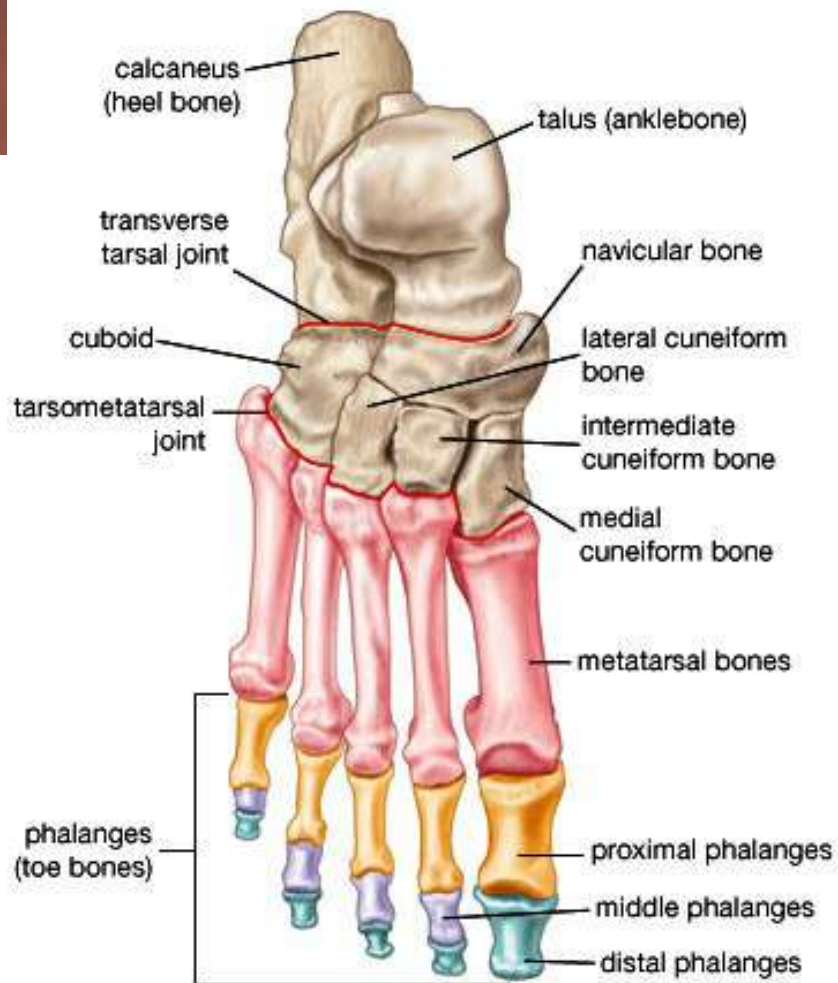
- **Cuboid**: has a deep groove on its inferior aspect. Is the most lateral in distal row.
- **Three cuneiform**: 3 small wedge-shaped bones (medial, intermediate & lateral). Articulate proximally with navicular & distally with first 3 metatarsal bones.

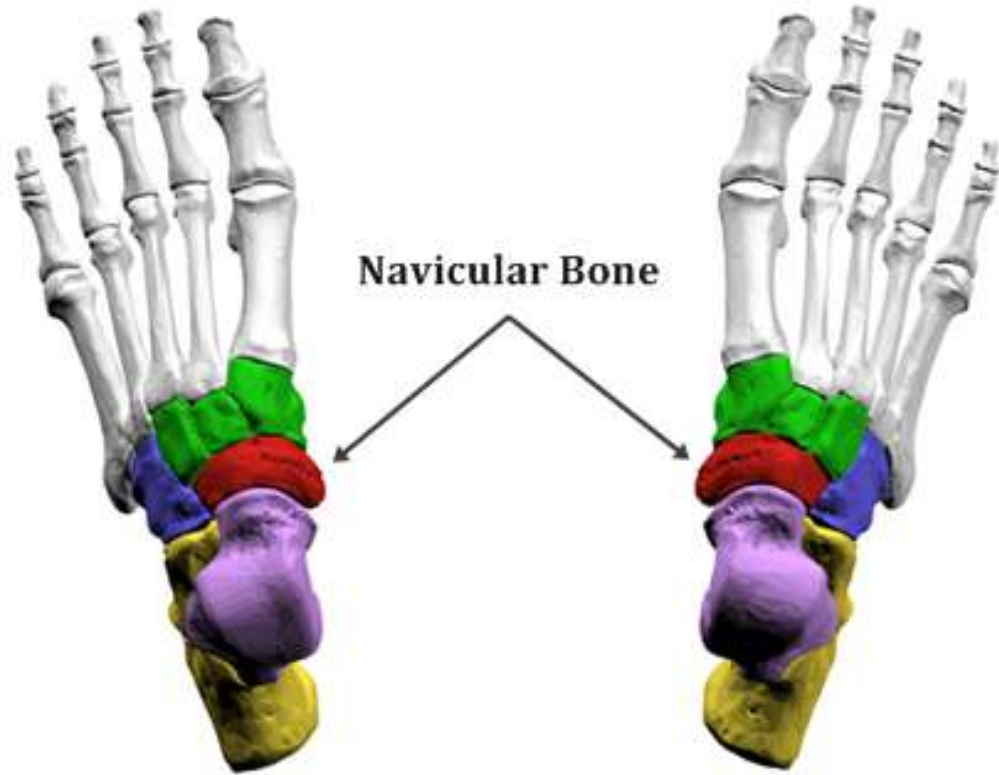
2- Metatarsal bones

- 5 in number
- Connecting tarsus to phalanges.
- Numbered from **medial side**.
- They have; base (proximal), shaft and head (distal).
- The 1st metatarsal is large & strong, supporting body weight. Its head is grooved on its inferior aspect by the medial and lateral sesamoid bone in the tendon of flexor hallucis brevis.

3- Phalanges

- Each toe has three phalanges (proximal, middle & distal) Except the big toe which has only two (proximal & distal).
- Each phalanx has base, shaft and head.





Arches of the foot

Three arches for foot

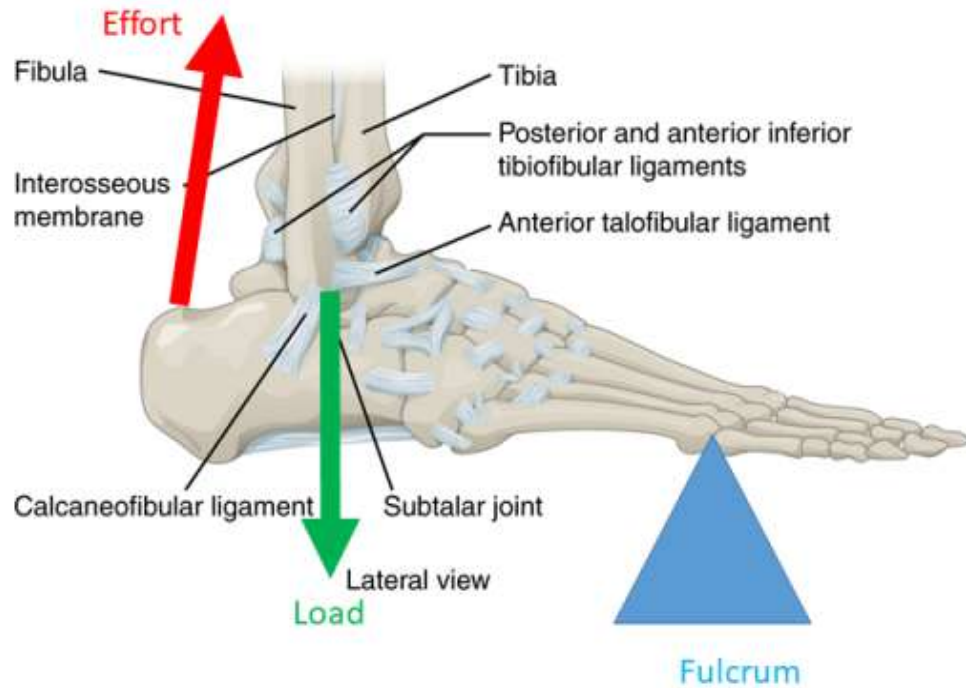
(1) Transvers arch

(2) longitudinal arches (medial & lateral).

Two important functions for foot;

1- support body weight 2- serve as a lever for propulsive forward body movement in walking & running.

- **Foot is not a single bone but formed of a series of small bones, connected by joints.**
 - **A segmented structure can hold up weight only if it is built in the form of an arch.**
 - **Medial longitudinal arch is the largest and the most important clinically.**
 - **Shape of bones, strong ligaments, tone of muscles, are important factors in arch support.**
- +plantar aponeurosis



1) **Transverse arch** consists of bases of metatarsals bones, cuboidal and 3 cuneiforms.

2) **Lateral longitudinal arch** consists of ; calcaneum, cuboid & 4th & 5th metatarsals bones.

3) **Medial longitudinal arch** consists of ; calcaneum, talus, navicular, three cuneiform bones & the first 3 metatarsals.

talus; the concave proximal surface of the navicular bone

The cuboid is the keystone.

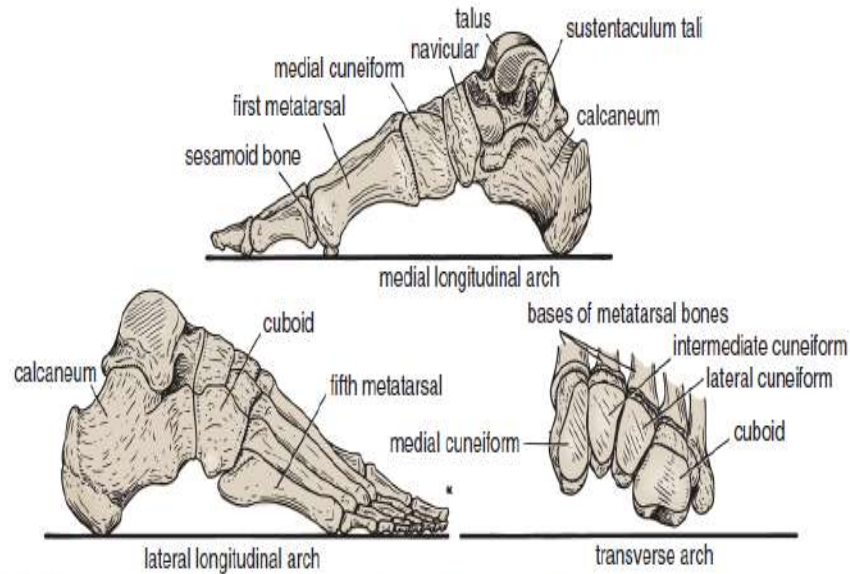
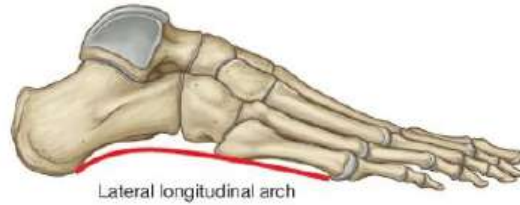
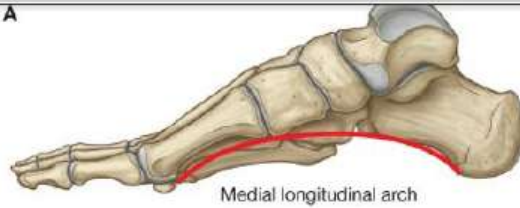


FIGURE 10.67 Bones forming the medial longitudinal, lateral longitudinal, and transverse arches of the right foot.

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- 1) **Transverse arch** consists of bases of metatarsals bones, cuboidal and 3 cuniforms.
- 2) **Lateral longitudinal arch** consists of ; calcaneum, cuboid & 4th & 5th metatarsals bones.
- 3) **Medial longitudinal arch** consists of ; calcaneum, talus, navicular, three cuniforms bones & the first 3 metatarsals.

A



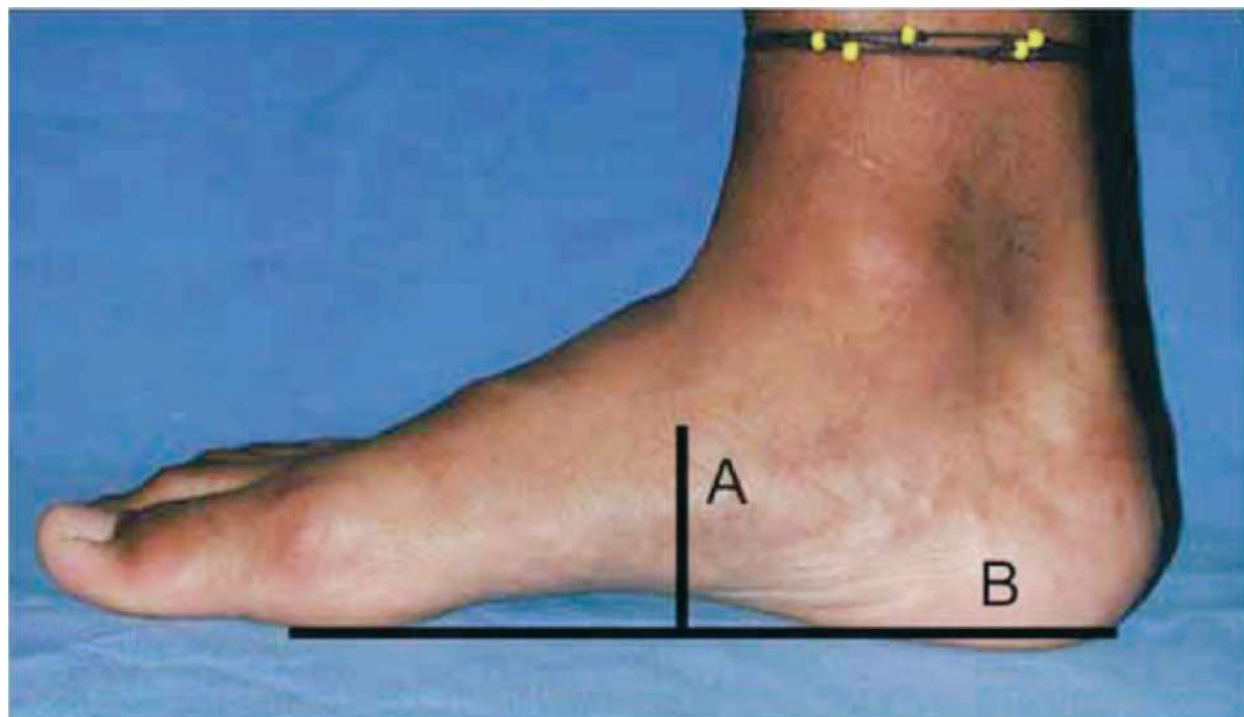
B



Drake: Gray's Anatomy for Students, 2nd Edition.

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Figure 6.107 Arches of the foot. **A.** Longitudinal arches, right foot. **B.** Transverse arch, left foot.





Pes planus



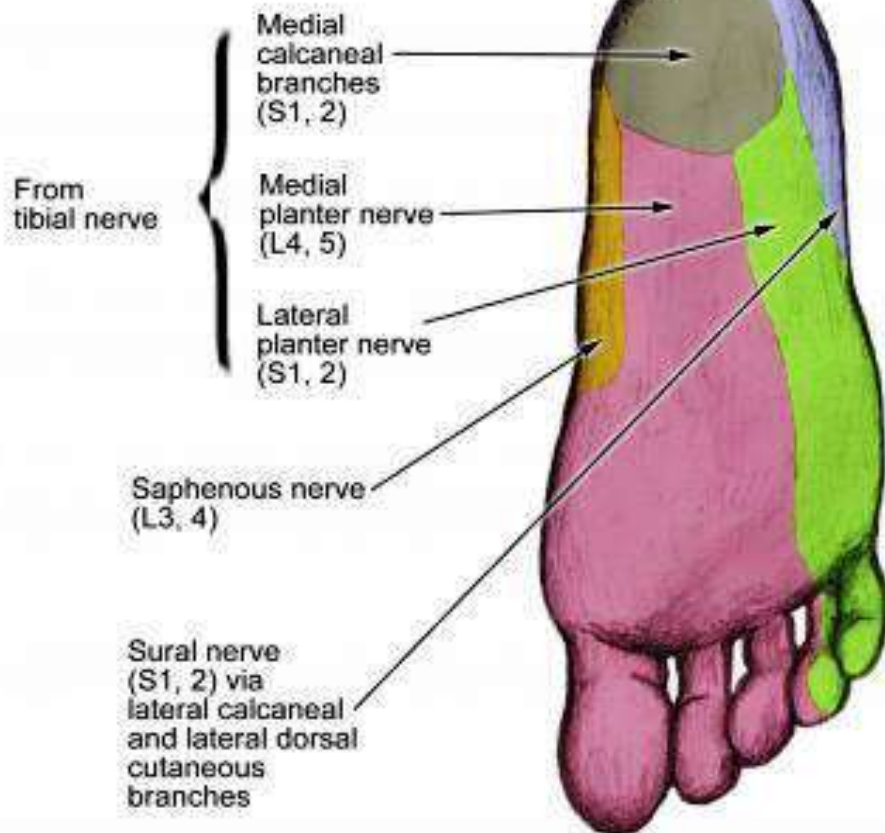
Pes cavum

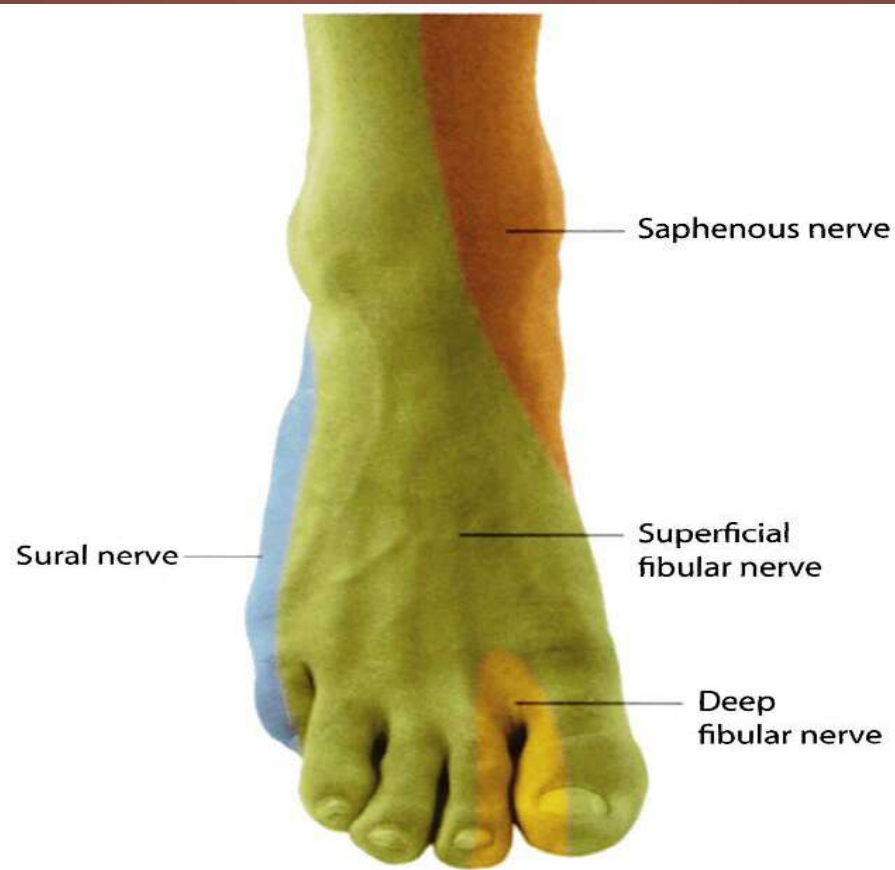
Sole of the foot

1) Skin

- Is thick & hairless
- It is firmly bound to deep fascia.
- Shows a few flexure creases.
- Sweat glands are present in large amount.
- Sensory nerve supply:
 - **Medial calcaneal branch** (innervates medial side of the heel)
 - **Medial plantar branch** (innervates medial 2/3 of sole.)
 - **Lateral plantar branch** (innervates lateral 1/3 of the sole)







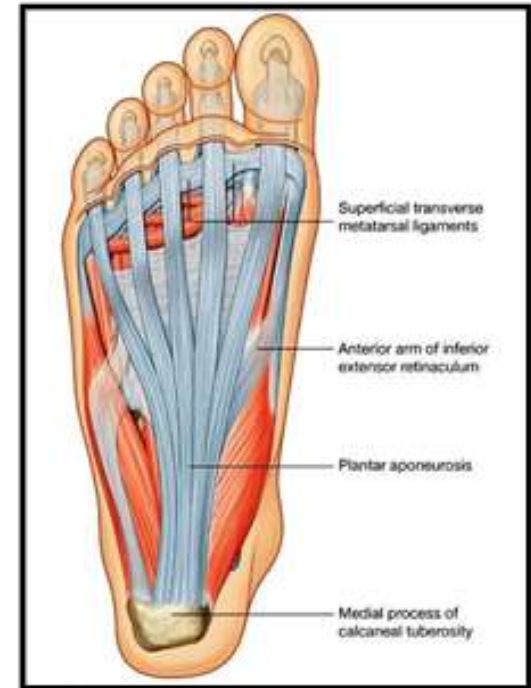
Cutaneous distribution

Cutaneous nerve	Origin	Area of skin supplied
1. Medial calcaneal branch	Tibial nerve.	Skin of the Medial side of the heel.
2. Branches from the Medial plantar nerve.	Terminal branch of Tibial nerve.	Skin of the Medial 2/3 of the sole & Medial 3 and ½ toes.
3. Branches from the Lateral plantar nerve.	Terminal branch of Tibial nerve.	Skin of the Lateral 1/3 of the sole & Lateral 1 and ½ toes.
4. Saphenous nerve	branch of femoral nerve	To skin of the Medial part of dorsum of foot as far as metatarsophalangeal joint of big toe & Medial border of the foot .
5. Deep peroneal nerve	Terminal branch of common peroneal nerve	To skin of the adjacent sides of the big and second toe of dorsum of the foot.
6. Superficial peroneal nerve	Terminal branch of common peroneal nerve	To skin of the greater intermediate part of the dorsum of the foot , medial side of the big toe & adjacent sides of the 2nd,3rd,4th & 5th toes.
7. Sural nerve	Branch of tibial nerve	To skin of lateral part of the dorsum of the foot, lateral border of the foot & lateral side of the little toe.

2)Deep fascia

■ Plantar aponeurosis

- Is triangular and occupies the central area of the sole.
- Is divided at the bases of the toes into 5 slips.
- Its apex is attached to calcaneum.
- **Its function:** is to give firm attachment to the overlying skin, to protect underlying vessels , nerves, tendons & to assist in maintaining arches of the foot



Muscles of the sole of the foot

(3,4,3,3)

- Are conventionally described in 4 layers from the superficial to deep layer.
- These layers have no significance beyond that of description & they represent the order in which the structures are uncovered and the general depth at which they lie.

- Sole muscles are chiefly concerned with supporting the arches of the foot.

First layer:

1. **Abductor** hallucis MPN
2. **Flexor** digitorum brevis MPN
3. **Abductor** digiti minimi

Second layer:

1. Quadratus plantae
2. Lumbricals (1st :MPN)
3. Flexor digitorum longus tendon
4. Flexor hallucis longus tendon

Third layer:

1. **Flexor** hallucis brevis MPN
2. **Adductor** hallucis
3. **Flexor** digiti minimi brevis

Fourth layer

1. Interossei
2. Peroneus longus tendon
3. Tibialis posterior tendon

First layer:

1. **Abductor hallucis**
2. **Flexor digitorum brevis**
3. **Abductor digiti minimi**

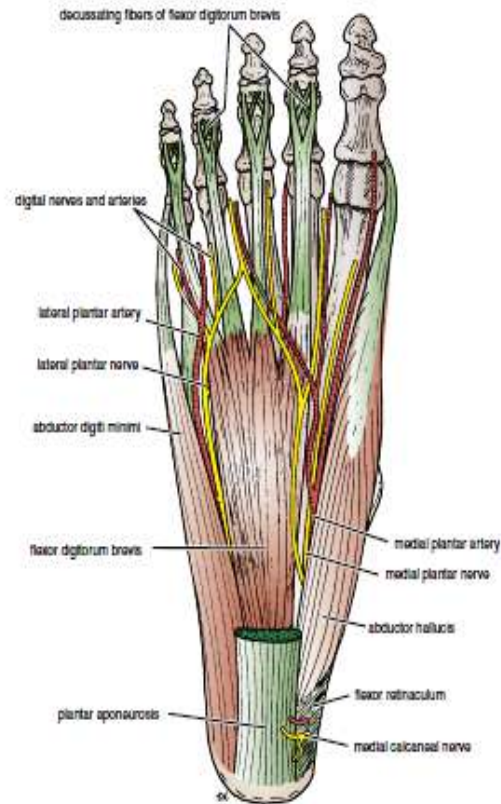
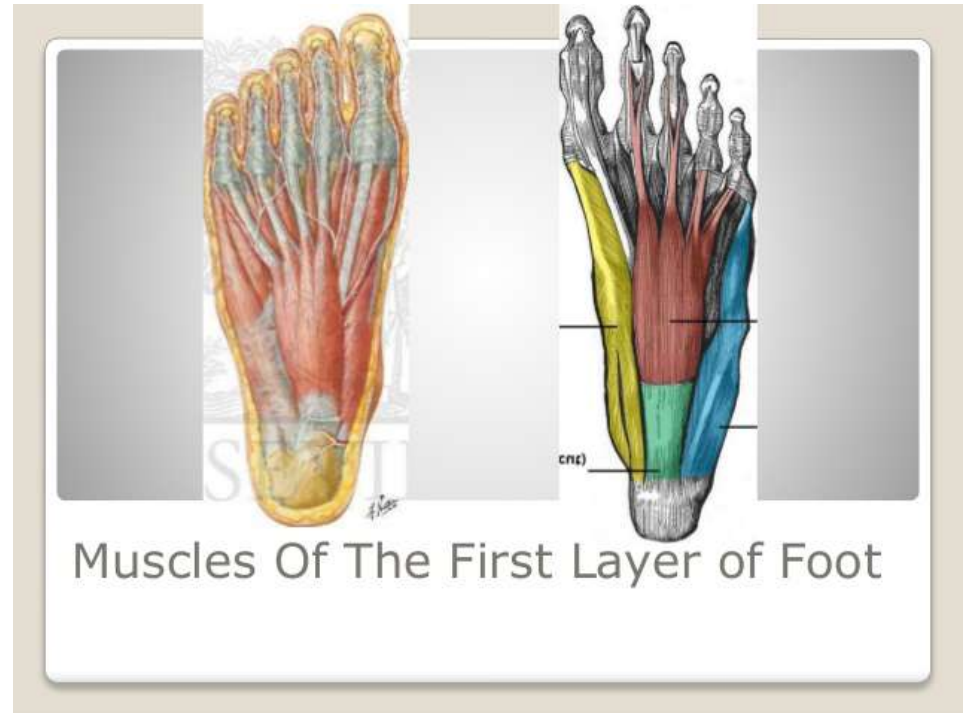


FIGURE 10.55 First layer of the plantar muscles of the right foot. Medial and lateral plantar arteries and nerves are also shown.



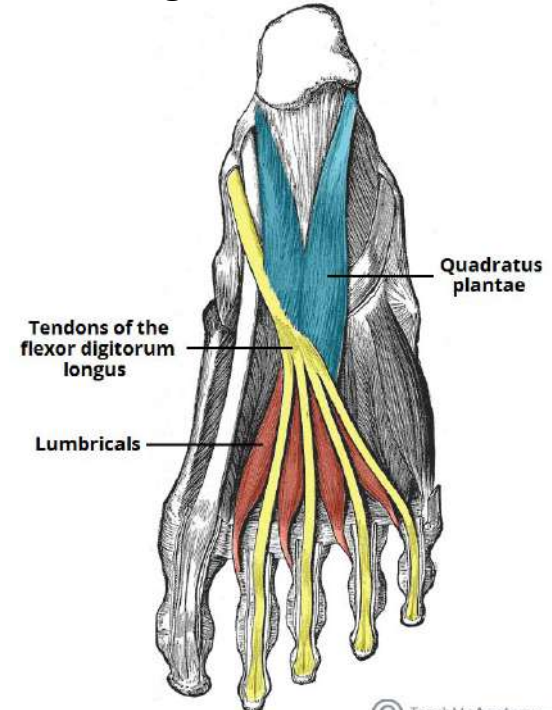
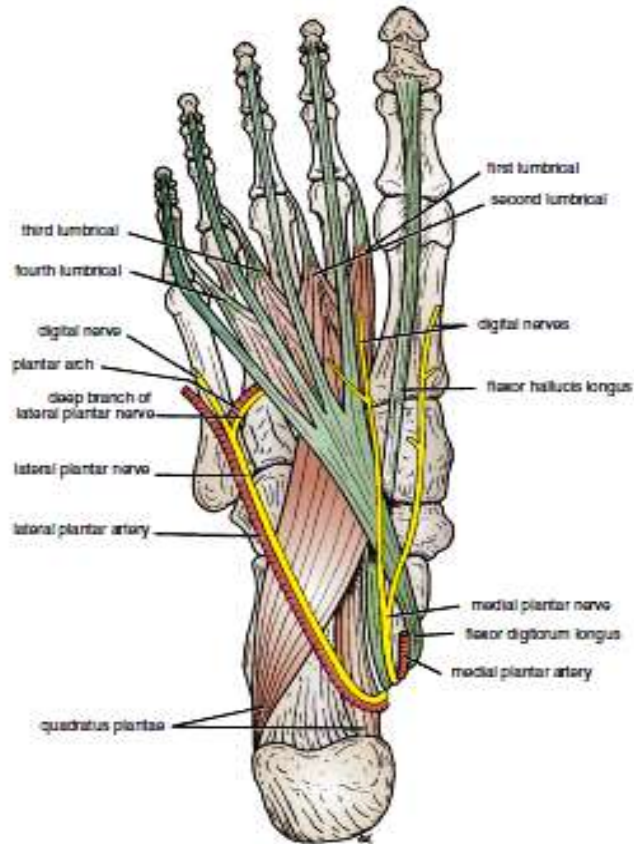
Second layer:

1. Quadratus plantae

2. Lumbricals

3. Flexor digitorum longus tendon

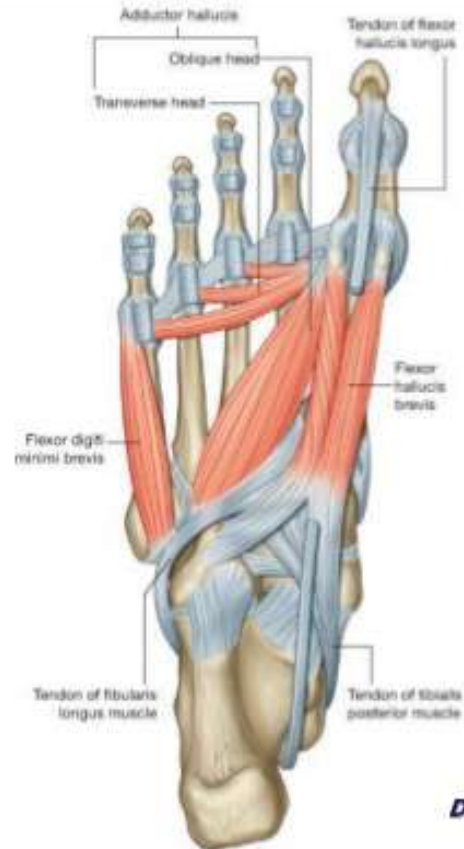
4. Flexor hallucis longus tendon

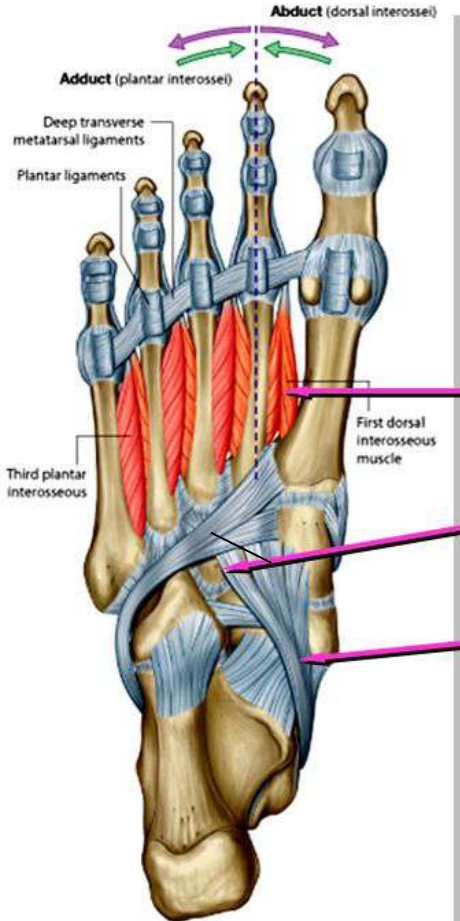


3RD LAYER

■ **Three Muscles:**

- 1) Flexor hallucis brevis.
- 2) Adductor hallucis
- 3) Flexor digiti minimi brevis

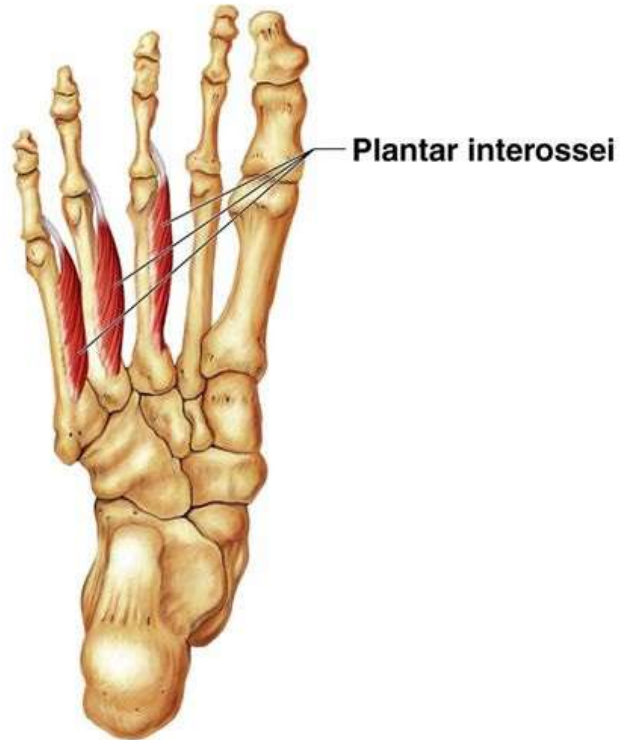




Fourth Layer

1. Interossei, (3 plantar + 4 dorsal).
2. Peroneus longus tendon,
3. Tibialis posterior tendon

Intrinsic Muscles of the Foot



(d) Fourth layer: plantar interossei



(e) Fourth layer: dorsal interossei

First layer of muscles in the sole of the foot

(spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Abductor hallucis	1 Medial process of calcaneal tuberosity	Medial side of base of proximal phalanx of great toe	Medial plantar nerve from the tibial nerve [S1, S2, S3]	Abducts and flexes great toe at metatarsophalangeal joint
Flexor digitorum brevis	2 Medial process of calcaneal tuberosity and plantar aponeurosis	Sides of plantar surface of middle phalanges of lateral four toes	Medial plantar nerve from the tibial nerve [S1, S2, S3]	Flexes lateral four toes at proximal interphalangeal joint
Abductor digiti minimi	3 Lateral and medial processes of calcaneal tuberosity, and band of connective tissue connecting calcaneus with base of metatarsal V	Lateral side of base of proximal phalanx of little toe	Lateral plantar nerve from the tibial nerve [S1, S2, S3]	Abducts little toe at the metatarsophalangeal joint

(*) - Abductors in 1st layer of Sole.

Second layer of muscles in the sole of the foot

Quadratus plantae	4 Medial surface of calcaneus and lateral process of calcaneal tuberosity	Lateral side of <u>tendon of flexor digitorum longus</u> in proximal sole of the foot	Lateral plantar nerve from tibial nerve [S1, S2, S3]	Assists flexor digitorum longus tendon in flexing toes II to V
Lumbricals	5 First lumbrical—medial side of <u>tendon of flexor digitorum longus</u> , associated with toe II; second, third, and fourth lumbricals—adjacent surfaces of adjacent tendons of flexor digitorum longus	Medial free margins of extensor hoods of toes II to V	First lumbrical—medial plantar nerve from the tibial nerve; second, third, and fourth lumbricals—lateral plantar nerve from the tibial nerve [S2, S3]	Flexion of metatarsophalangeal joint and extension of interphalangeal joints (Same as Hand Lumbricals)

(*) - Lumbricals in 2nd Layer of Sole.

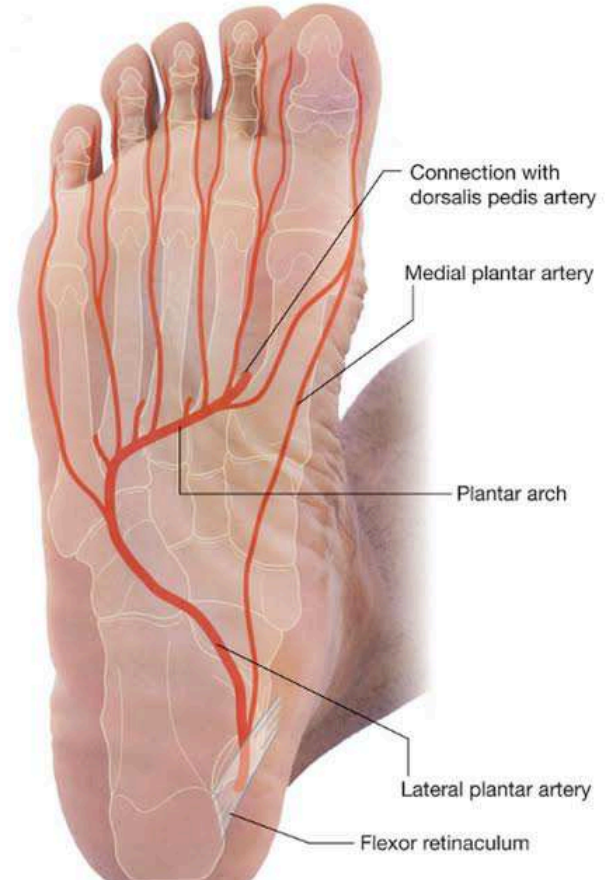


Muscle	Action
Abductor hallucis	Flexion & abduction of big toe
Flexor digitorum brevis	Flexion of lateral 4 toes
Abductor digitiminimi	Flexion & abduction of Of little toe.
Quadratus plantae	Assists flexor digitorum longus in flexing lateral 4 toes.
Lumbricals	Extends toes at interphalangeal joints
Flexor halluces brevis	Flexes metatarsophalangeal joints of big toe
Adductor hallucis	Flexes metatarsophalangeal joint of big toe
Flexor digitiminmi	Flexes metatarsophalangeal joint of little toe
Dorsal interossei	Abduction of toes, flexes metatarsophalangeal joints& extends interphalangeal joints.
Plantar interossei	Adduction of toes, flexes metatarsophalangeal

Arteries of the sole

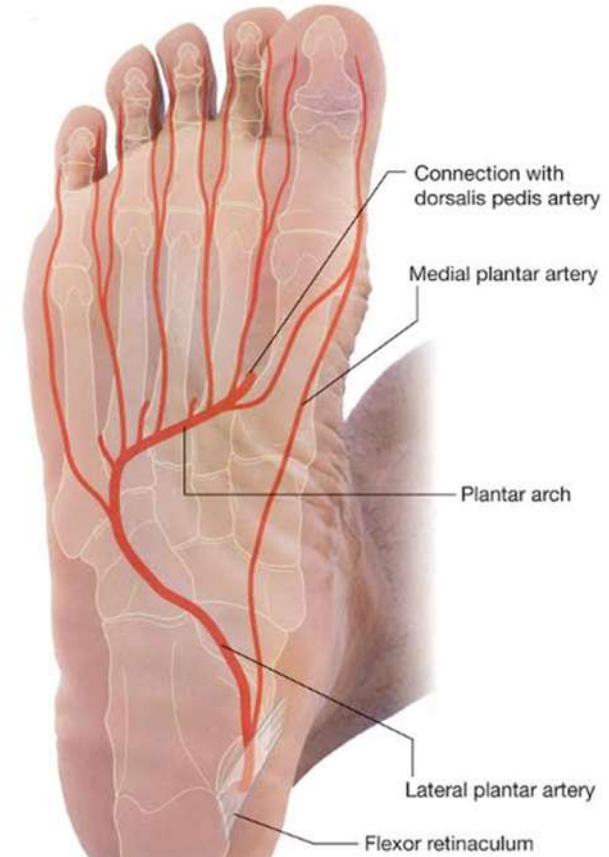
1) Medial plantar artery

- Is the smaller of the terminal branches of the **posterior Tibial artery**.
- It arises beneath to flexor retinaculum
- Runs with the medial plantar nerve.
- Ends by supplying the medial side of the big toe.
- It gives, muscular, cutaneous & articular branches.



2) Lateral plantar artery

- Is the larger of the terminal branches of the posterior Tibial artery.
- Runs with the lateral plantar nerve.
- It arises beneath to flexor retinaculum
 - On reaching the base of 5th metatarsal bone, it curves medially to form **the plantar arch**. **مهم جدا**
- Plantar arch gives off plantar digital arteries to the toes.



3) Dorsalis pedis artery (Dorsal artery of the foot)

- On entering the sole of the foot between the two heads of 1st dorsal interosseous muscle, dorsalis pedis artery immediately **joins lateral plantar artery.**

Branches: 1st plantar metatarsal artery which supplies cleft between big & second toes.

❑ Branches of dorsalis pedis artery :

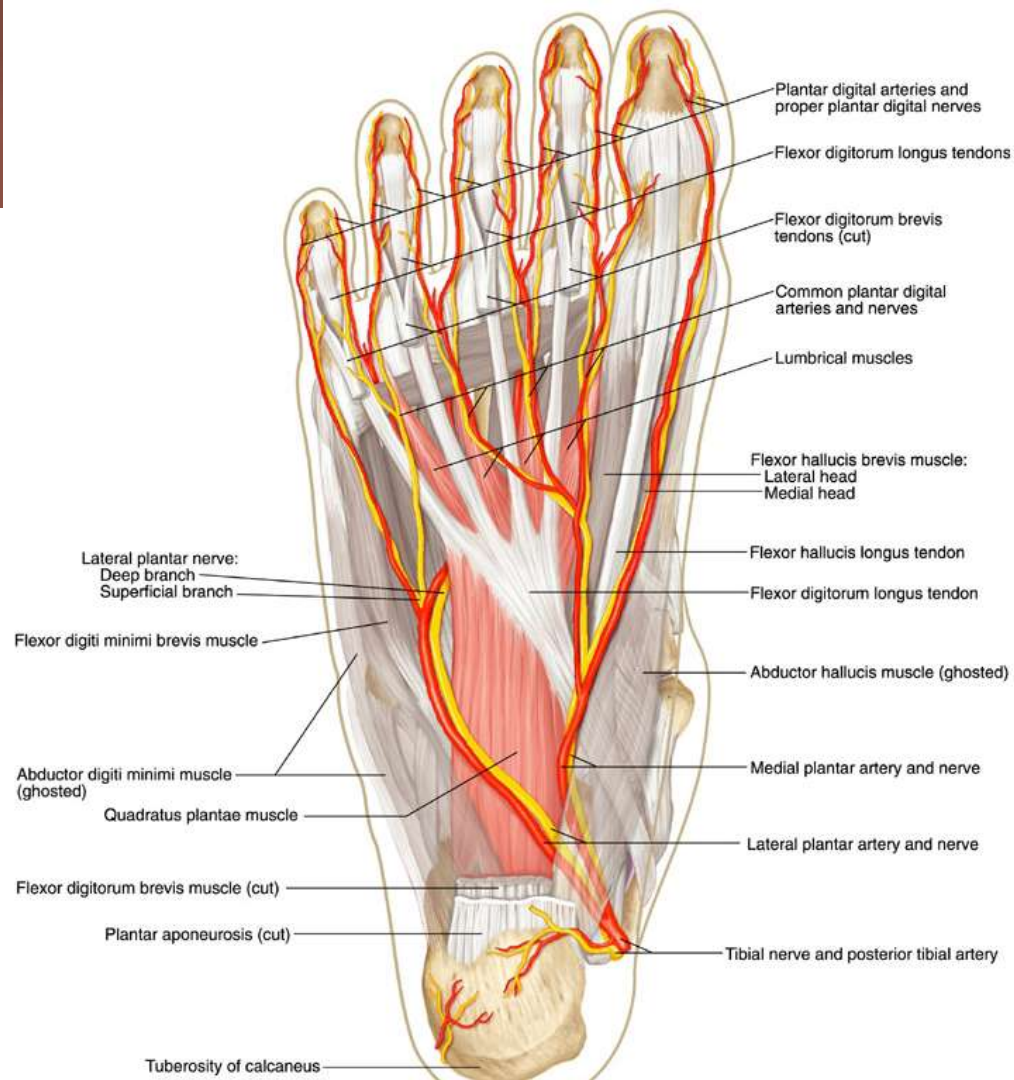
1-lateral tarsal artery

2-Medial tarsal arteries.

3-Arcuate artery.

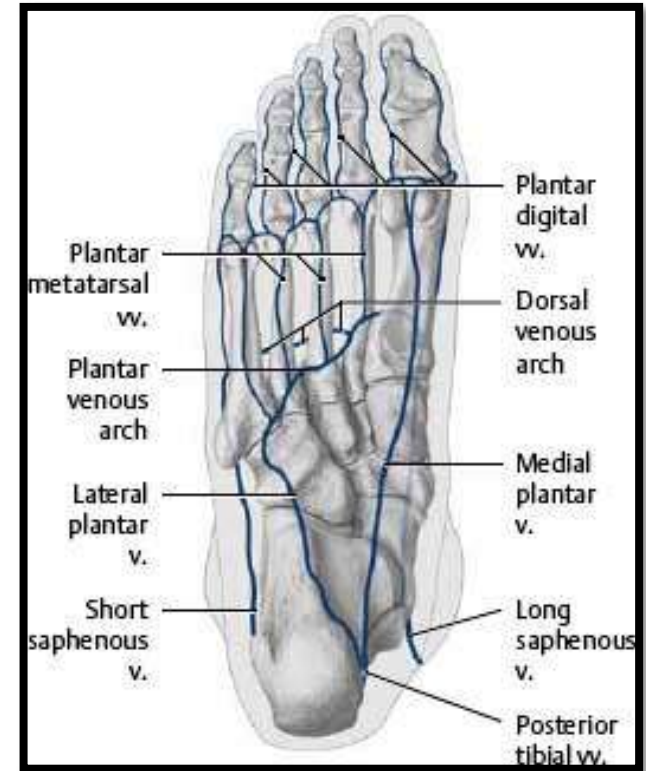
4-First dorsal metatarsal artery.

5-First plantar metatarsal artery.



Veins of the sole of the foot

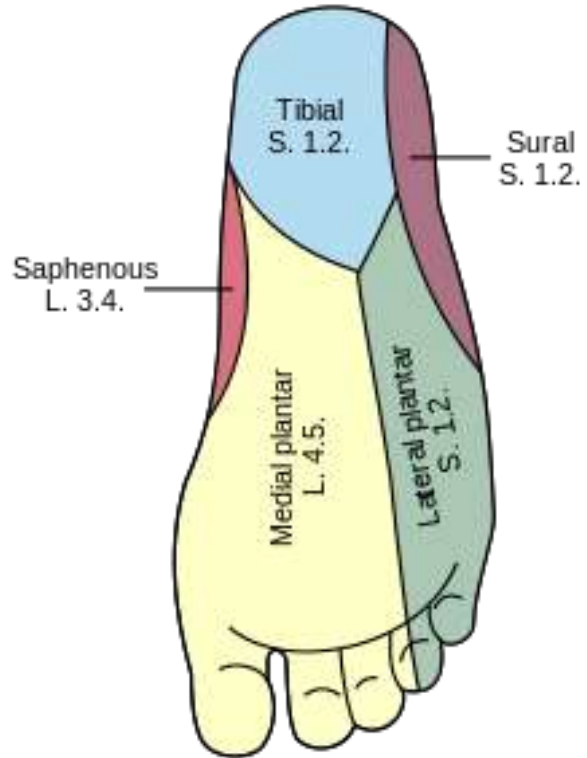
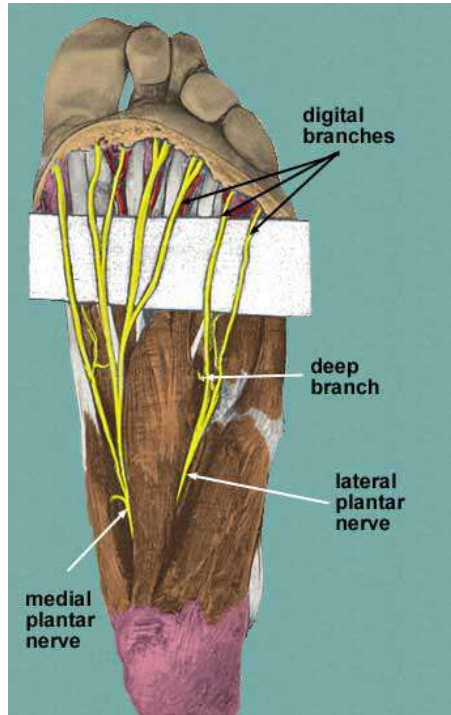
- Medial & lateral plantar veins accompany medial & lateral plantar arteries.
- End: by uniting behind the medial malleolus to form posterior tibial venae comitantes.



Nerves of the sole of the foot

1) Medial plantar nerve

- Is the terminal branch of tibial nerve.
- **Branches:**
 - 1-Muscular: to abductor hallucis, flexor digitorum brevis, flexor hallucis brevis & the first lumbrical muscle.
 - 2-Cutaneous branches:
 - Run to sides of the medial three and one-half toes, it also supplies their nail beds and tips of the toes.



2- Lateral plantar nerve

- Is the terminal branch of tibial nerve.
- It divides into superficial & deep.

Branches: from the main trunk, from superficial terminal branch and from deep terminal branch.

- Branches:

- From the main trunk:

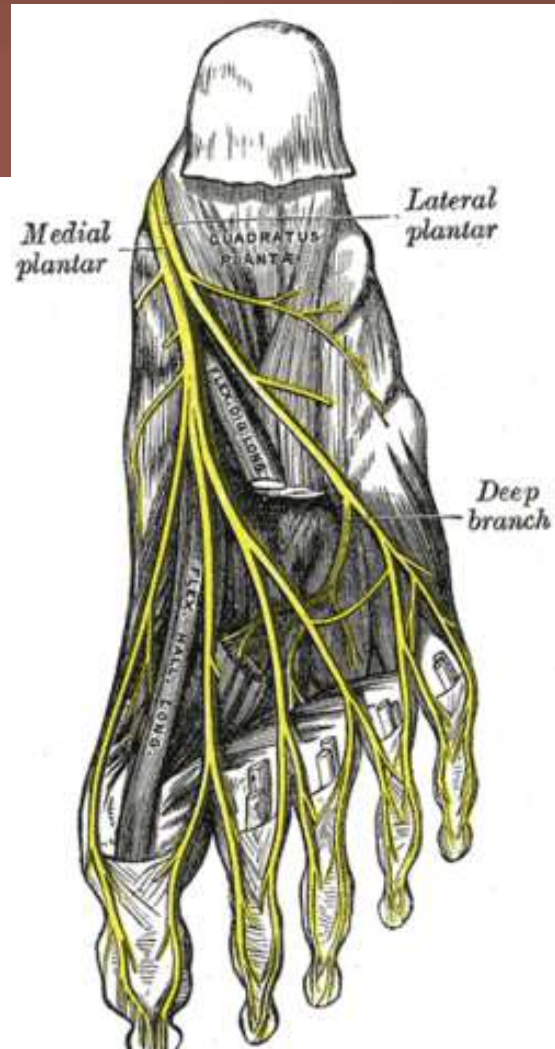
To quadratus
plantae&abductor digiti
minimi, cutaneous to the
lateral part of the sole.

- From the superficial branch:

To rest of muscles+ sides of the
lateral one and one-half toes,
extends to nail beds and tips of
the toes.

- From Deep branch:

Muscular: supplies adductor
hallucis, 2nd, 3rd, 4th lumbricals
and all the interossei.



- **Medial plantar nerve** in the sole of foot supplies abductor hallucis, flexor digitorum brevis, flexor hallucis brevis & the **first** lumbrical muscle.

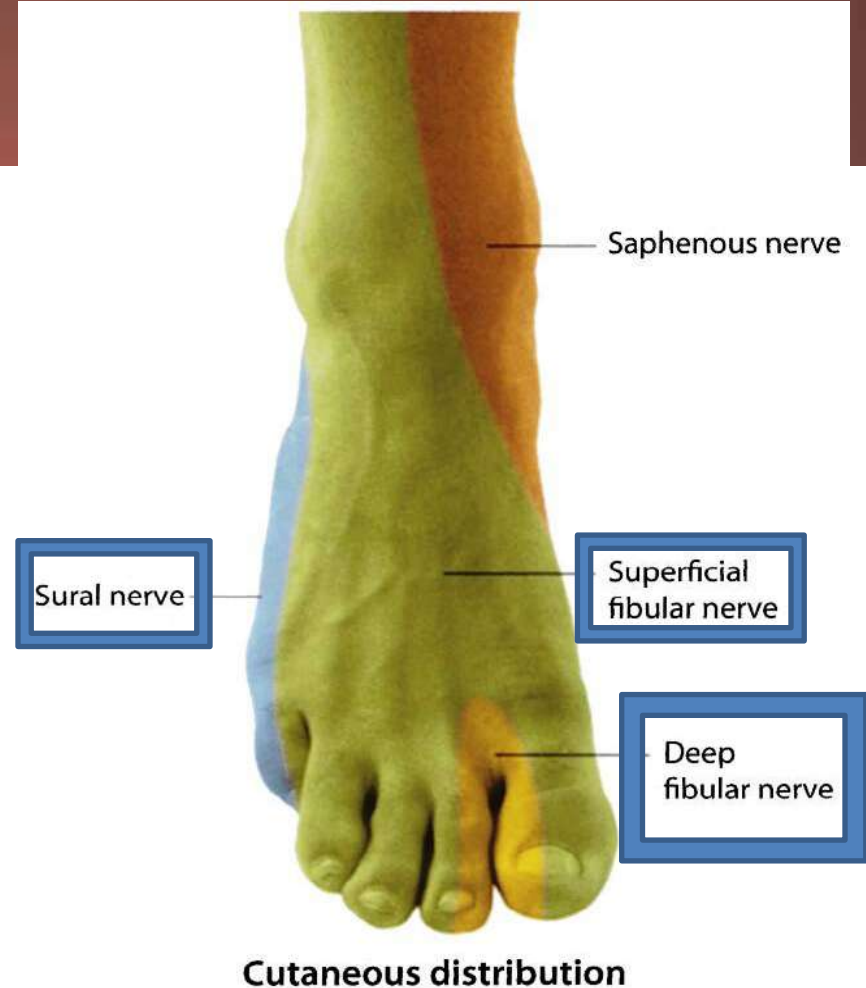
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The dorsum of the foot



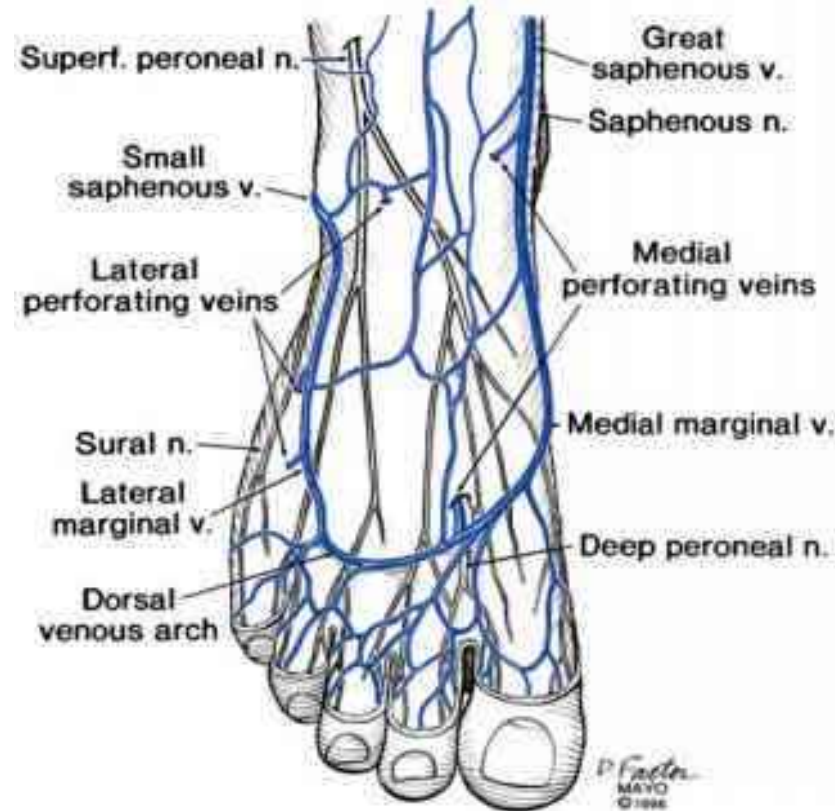
Skin :

- Is thin, hairy and freely mobile on underlying tendons & bones.
- Sensory innervation:
Superficial peroneal, deep peroneal , saphenous and sural nerves.



Dorsal venous arch(network):

- Dorsal venous network lies in subcutaneous tissue & drains on the medial side into great saphenous vein & on the lateral side into the small saphenous vein. 



Muscles on the dorsum of the foot

Extensor Digitorum Brevis:

O: from ant. calcaneum & inferior extensor retinaculum.

I: It gives 4 tendons:

The most medial tendon (tendon of extensor hallucis brevis):
inserted into base of proximal phalanx of the big toe.

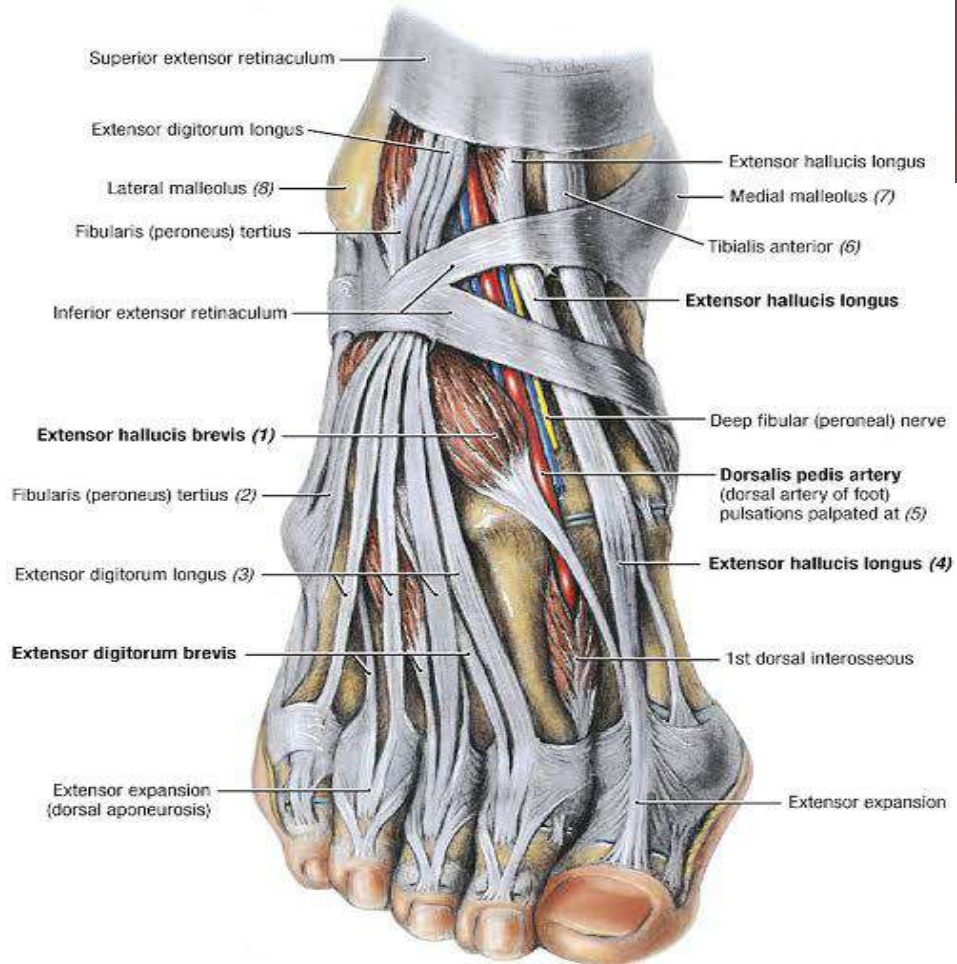
The lateral 3 tendons: join the long extensor tendons passing to 2nd, 3rd and 4th toes.

NS: terminal part of **deep peroneal nerve**

A: Extends the 1st, 2nd, 3rd, 4th toes at interphalangeal & metatarsophalangeal joints.



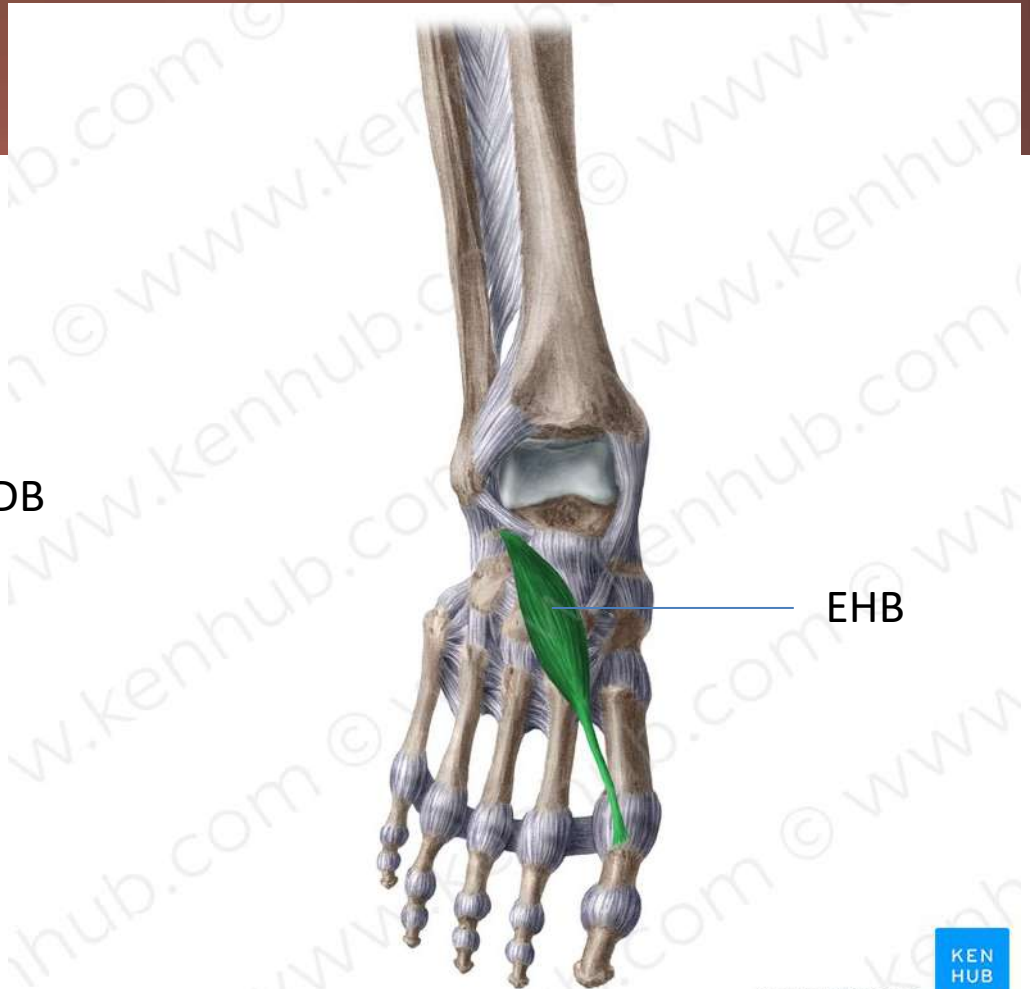
A. Superior View



B. Superior View



EDB

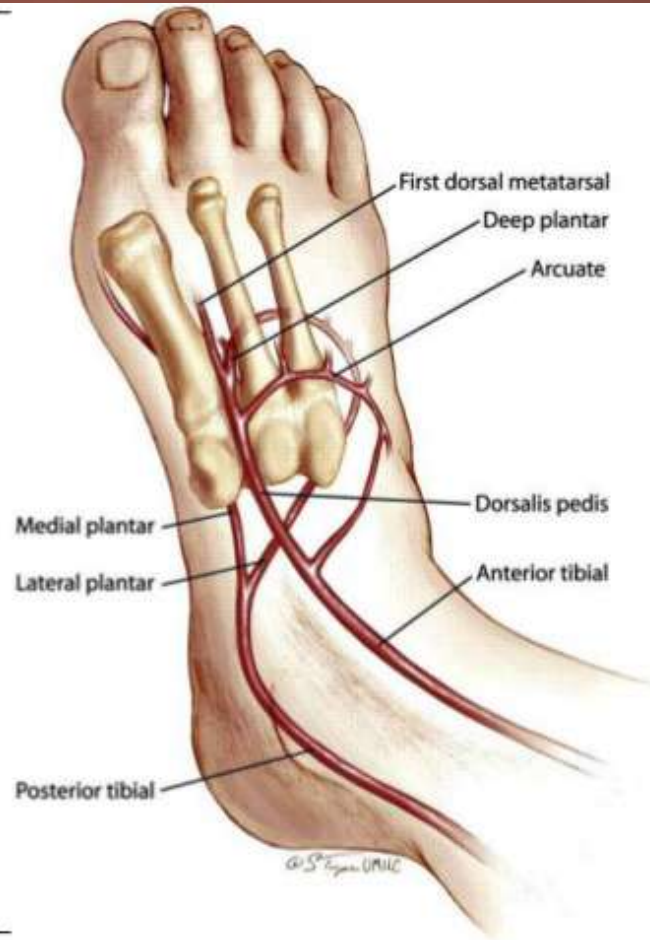


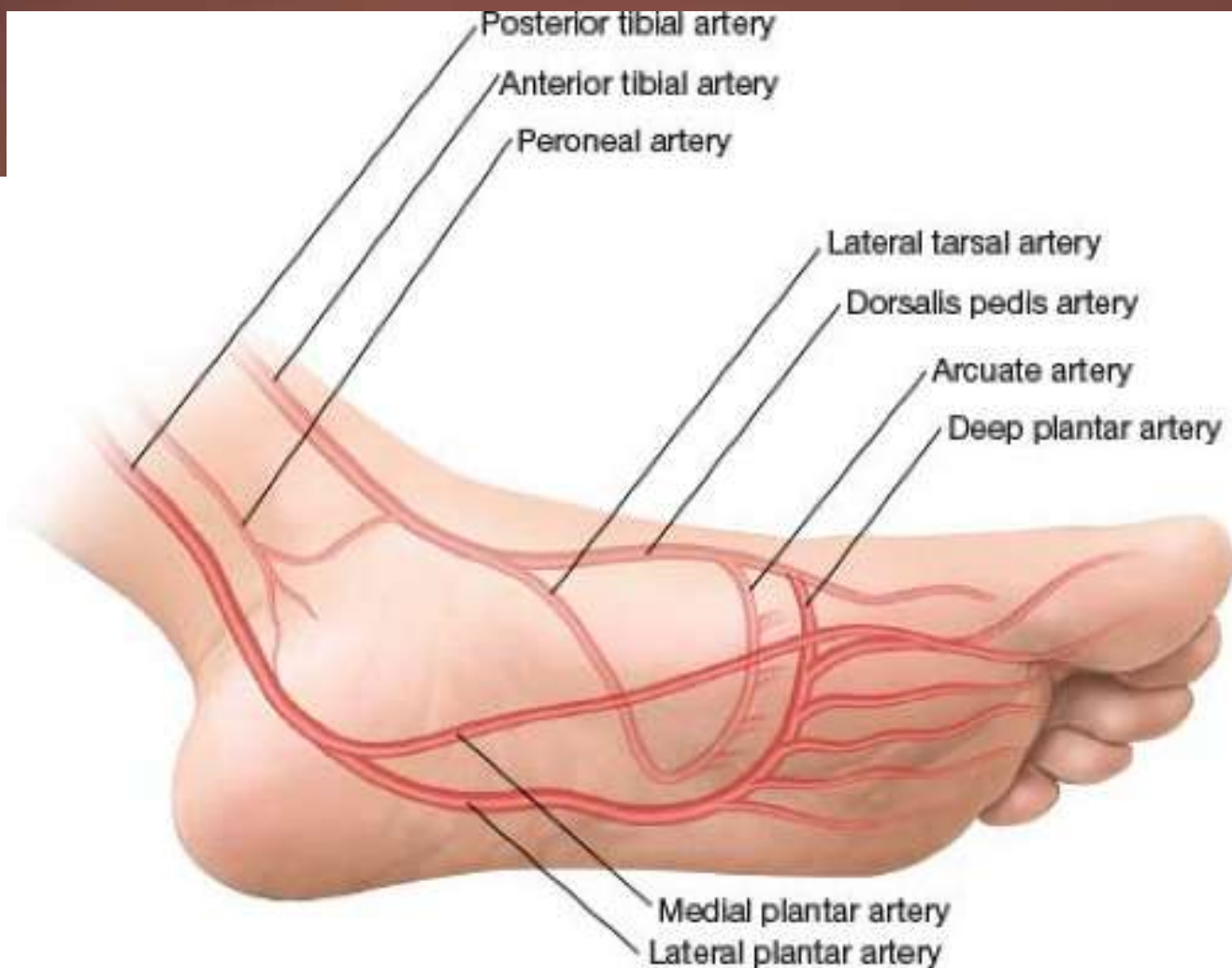
EHB

Artery of the dorsum of the foot

Dorsalis pedis artery

- Begins in the front of the ankle as a continuation of anterior tibial artery.
- Ends by passing to the sole, where it joins the lateral plantar artery & completes the plantar arch
- Its pulsation can be easily felt between the tendons of Extensor digitorum longus and Extensor hallucis longus.



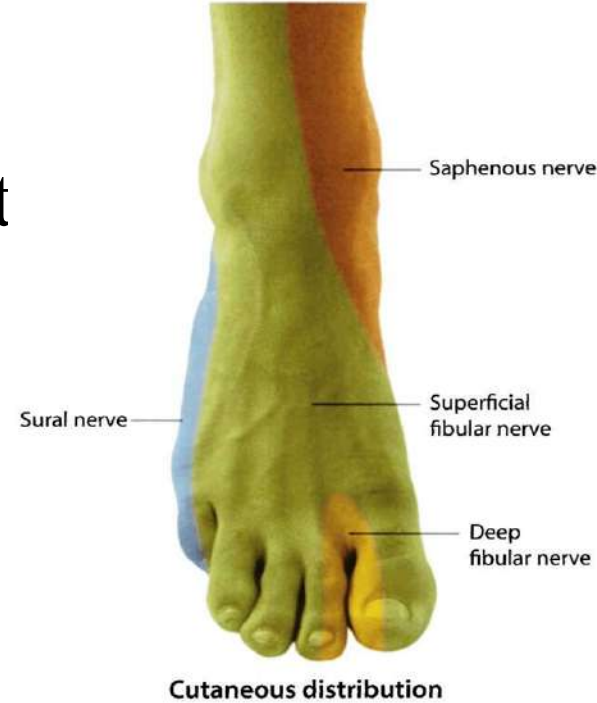


Nerve of the dorsum of the foot

Deep peroneal nerve:

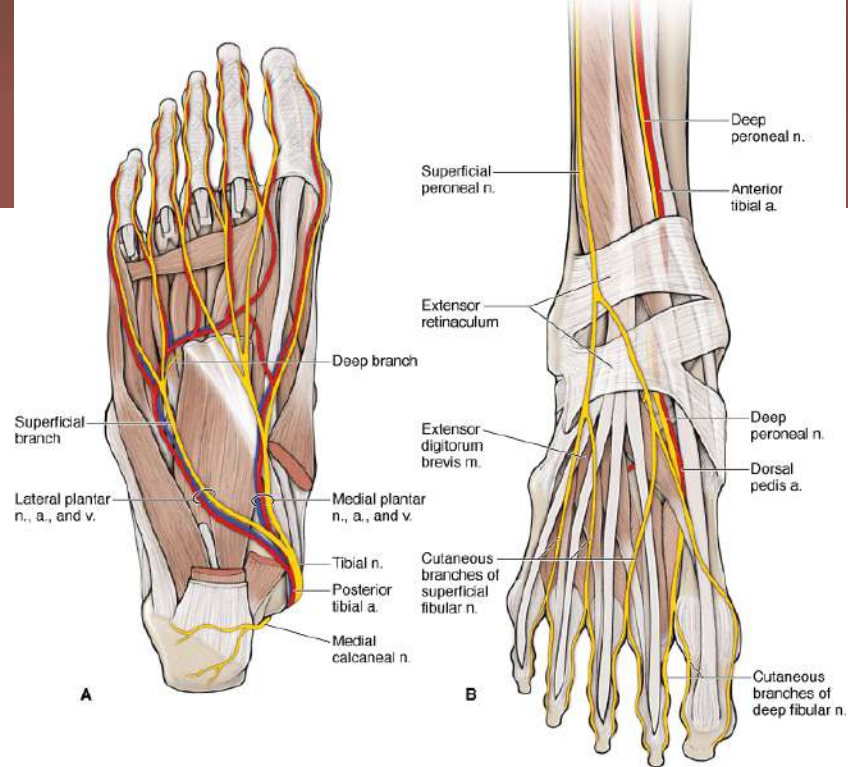
- ❖ It is one of the terminal branches of **common peroneal nerve**.
- ❖ It enters the dorsum of the foot by passing deep to **extensor retinaculum**, on the lateral side of dorsalis pedis artery.

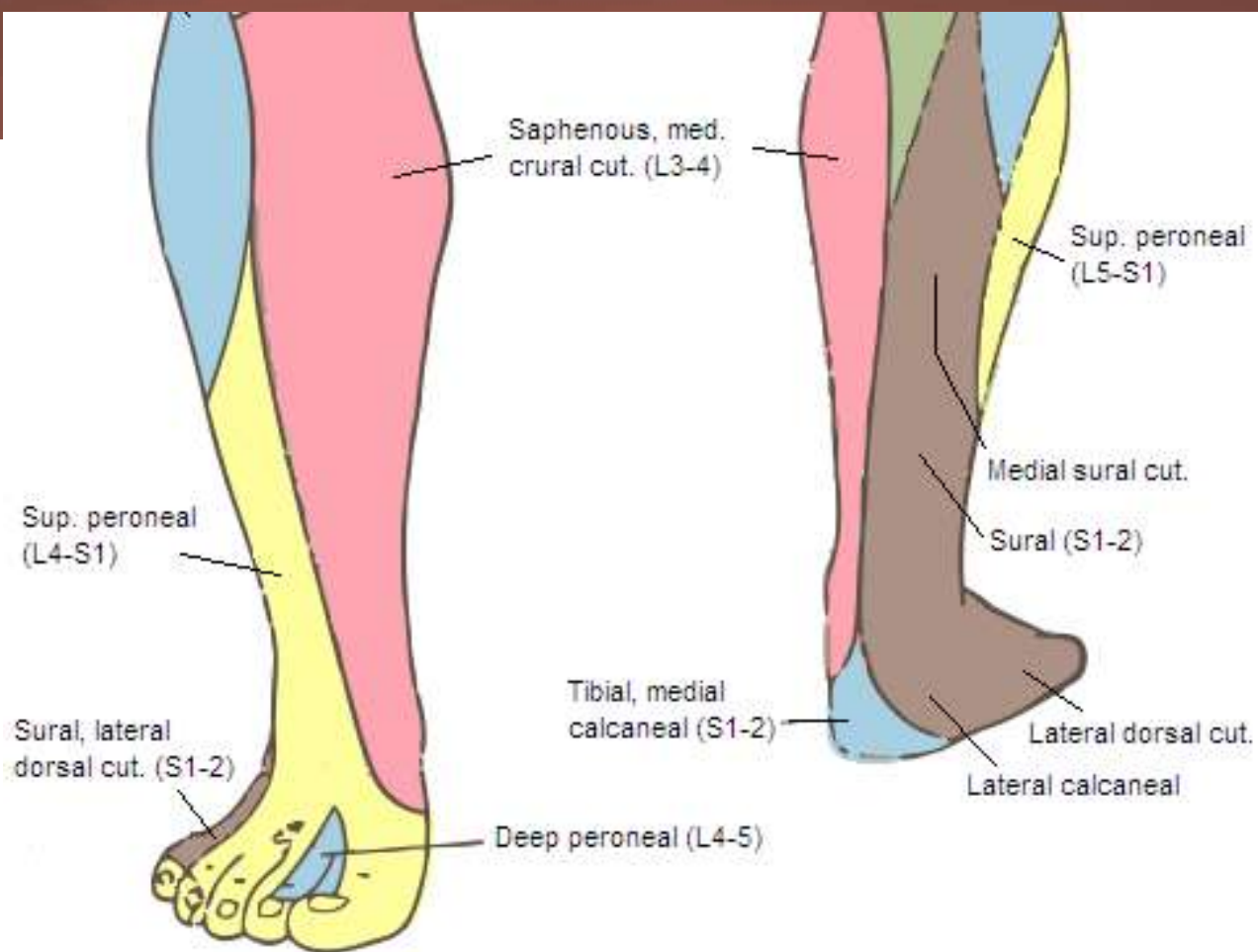
Branches: muscular (to extensor digitorum brevis) , cutaneous (adjacent side of big and 2nd toes) & articular (to joints of foot).



Other sensory nerves for the dorsum of the foot:

- ❖ **Superficial peroneal (common peroneal nerve).**
- ❖ **Sural nerve (tibial nerve).**
- ❖ **Saphenous nerve (femoral nerve).**









Congratulation
Med2023 for ending
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the course of the
LOWER LIMB